



BHARATHIDASAN INSTITUTE OF MANAGEMENT

TIRUCHIRAPPALLI

influencing Tomorrow

Advanced Predictive Analytics

Module 2 – Advanced Regression & Model Interpretability

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Agenda – Module 2: Advanced Regression & Model Interpretability

2.1. Advanced Regression Techniques

- Linear/Logistic Regression Advanced Methods:
 - Regularization (Lasso, Ridge, ElasticNet)
 - Generalized Linear Models (Poisson, Gamma regression)
 - Survival Analysis (Kaplan-Meier, Cox Proportional Hazards)

2.2. Model Interpretability & Scalability

- SHAP/LIME for explaining black-box models.
- Business storytelling with regression outputs.

2.3. Neural Network

Advanced Regression Techniques



Advanced Regression Techniques

Limitations of Basic Regression:

Multicollinearity, overfitting, high-dimensionality issues

Business example: Predicting customer lifetime value (CLV) with 100+ features

Why we need Advanced Regression Techniques –

Traditional regression assumptions and their fragility in real business data.

- Real examples:
 - Marketing spend vs. sales → multicollinearity.
 - Customer churn → imbalance.
 - Insurance claim cost → skewed distributions.
- Why we need “advanced regression”:

Robustness + Scalability + Interpretability



Topic 1: Regularization Techniques (Lasso, Ridge, ElasticNet)

Regularization means **adding a penalty for model complexity** to avoid overfitting.

Think of it as telling your model:

“Don’t try to memorize the training data—focus on the general trend.”

- Regularization = penalty-based complexity control.
- Key Intuition: Adds constraint to minimize overfitting in high-dimensional spaces.
- Visuals: Contour plots of Lasso vs. Ridge penalty.



Regularization Technique - Comparison Table

Technique	Penalty	Feature Selection	Handles Multicollinearity	Stability	Business Use	Strength	Weakness	Best For
Lasso (L1)	Sum of abs(coeffs)	Yes (forces zero)	Partial	Can drop correlated vars	Marketing driver selection	Feature selection, sparse model	May drop important correlated vars	Marketing attribution
Ridge (L2)	Sum of squares	No	Excellent	Stable coefficients	Credit scoring, risk models	Stable, smooth	No variable elimination	Credit risk, demand forecasting
ElasticNet	Mix ($\alpha \times L1 + \beta \times L2$)	Partial	Good	Balanced	Price elasticity modeling	Balanced control	More tuning	High-dim datasets (customer analytics)



Regularization Technique - Mathematical Intuition

In OLS regression, we minimize:

$$RSS = \sum (y_i - \hat{y}_i)^2$$

In **Ridge** (L2 regularization):

$$\text{Minimize } RSS + \lambda \sum \beta_j^2$$

→ discourages large coefficients, keeps all variables.

In **Lasso** (L1 regularization):

$$\text{Minimize } RSS + \lambda \sum |\beta_j|$$

→ forces some coefficients to zero → automatic feature selection.

ElasticNet = mix of both (balances stability + sparsity).



Regularization Technique - Practical Challenges & Resolutions

Real-World Example:

- Case:** Predicting a customer’s annual spend based on 100 marketing variables.
- Lasso → Keeps only the top 10 strongest drivers.
 - Ridge → Keeps all 100 variables but reduces extreme effects.
 - ElasticNet → Keeps ~30 with balanced weights.

Challenge	Description	Resolution Strategy
Multicollinearity	Coefficients unstable in OLS	Ridge / ElasticNet regularization
Feature Overload	Too many predictors (e.g., marketing campaigns)	Use Lasso to shrink irrelevant features
Scaling Required	Regularization sensitive to units	Standardize variables before fitting
Interpretability	Coefficients penalized and distorted	Use standardized coefficients and SHAP post-hoc explanations



Regularization Technique - Python Lab Snippet

(Lasso, Ridge comparison; include cross-validation)

```
from sklearn.linear_model import RidgeCV, LassoCV, ElasticNetCV
ridge = RidgeCV(alphas=[0.1,1,10]).fit(X_train, y_train)
lasso = LassoCV(alphas=[0.001,0.01,0.1,1]).fit(X_train, y_train)
elastic = ElasticNetCV(l1_ratio=[.2,.5,.8]).fit(X_train, y_train)
print("Best alpha Ridge:", ridge.alpha_, "R2:", ridge.score(X_test, y_test))
print("Best alpha Lasso:", lasso.alpha_, "R2:", lasso.score(X_test, y_test))
print("Best alpha elastic:", elastic.alpha_, "R2:", elastic.score(X_test, y_test))
```

Exercise: Choose optimal alpha via grid search, interpret top-5 coefficients, narrate their business meaning.

Hint: Compare their R^2 scores and top features.

Topic 2: Generalized Linear Models (GLMs)

Linear regression assumes:

- Response (Y) is continuous
- Residuals are normally distributed

But in reality:

- Y can be **counts (Poisson)** → number of claims
- Y can be **skewed positive (Gamma)** → insurance cost
- Y can be **binary (Logistic)** → churn, loan approval

GLMs extend regression by linking the mean of Y to predictors via a **link function**.

GLM - Concept

- Regression for non-normal targets – basically it extends linear regression for non-normal distributions
- Components: Random component + Systematic component + Link function
- Types:
 - Poisson (count data) - Count outcomes (e.g., number of transactions)
 - Gamma (skewed continuous) - Skewed positive outcomes (e.g., insurance claims)
 - Logistic (binary outcomes) - Uses link functions (log, logit, inverse)



GLMs Technique - Comparison Table

Key Equations (Conceptual):

$$E[Y] = \mu, g(\mu) = X\beta$$

Where $g(\cdot)$ is the **link function** (log, logit, identity, etc.)

Model	Target Type	Link Function	Example Business Use	Key Limitation
Poisson	Count (0,1,2,3,...)	Log	# of store visits, claims	Overdispersion
Negative Binomial	Overdispersed count	Log	Retail transactions	Extra param tuning
Gamma	Continuous + skewed	Inverse or Log	Claim cost, sales	Sensitive to outliers
Logistic	Binary (0/1)	Logit	Churn prediction	No time component



GLMs Technique - Practical Challenges & Resolutions

Business Example

- Predicting number of claims per policyholder (Poisson)
- Forecasting energy usage or insurance premium amounts (Gamma)
- Predict whether claim will be approved (Logistic regression)

Challenge	Issue	Resolution
Overdispersion	Poisson variance $\neq$ mean	Use Negative Binomial
Zero Inflation	Excess zero counts	Zero-Inflated Poisson
Interpretability	Coefficients in log scale	Convert to incidence rate ratios / Exponentiate to rate ratio
Model Fit	Convergence issues	Check multicollinearity, scaling



GLMs Technique – Python Base

Poisson model for “number of transactions per customer”
Add comparison with OLS and visualize residuals.

```
import statsmodels.api as sm
import statsmodels.formula.api as smf
```

```
# Poisson Regression Example
df = pd.read_csv("claims_data.csv")
poisson_model = smf.glm(formula="claims ~ age + income +
policy_type",
                        data=df, family=sm.families.Poisson()).fit()
print(poisson_model.summary())
```



Topic 3: Survival Analysis



Used to predict “**time until an event occurs**”, e.g.:

- Customer churn
- Employee attrition
- Equipment failure

Unlike regression, it deals with **censored data** – we don’t always observe the event for everyone (e.g., still active customers).

Core Techniques:

• **Kaplan-Meier Curve** – Non-parametric survival probability: It estimates survival probability over time
→ “What % of customers stay after 6 months?”

• **Cox Proportional Hazards** – Effect of covariates on survival: It models effect of predictors on survival time
→ “How does low engagement affect churn rate?”

• **Monte Carlo Simulation** – simulate possible lifetimes: It is used for scenario testing (e.g., churn time forecasting)
→ “Forecast future churn distribution”



Survival Analysis - Comparison Table

Method	Nature	Inputs	Outputs	Use Case
Kaplan-Meier	Non-parametric	Time, Event	Survival curve	Churn visualization
Cox PH	Semi-parametric	Covariates	Hazard ratios	Churn driver analysis
Weibull / Exponential	Parametric	Time, Covariates	Predict lifetimes	Equipment failure



Survival Analysis - Practical Challenges & Resolutions

Business Example

- A telecom firm wants to know **how long a customer stays**.
- Kaplan-Meier → 70% retention after 12 months.
- Cox PH → Older customers 1.5× more likely to stay.

Challenge	Description	Solution	Fix
Censoring	Customers still active (no churn yet)	Proper censor flag handling	Include censored flag
Non-proportional hazards	Effect changes over time	Stratify / Time-varying covariates	Time-varying covariates
Interpretability	Hazard ratios complex	Convert to “Risk multipliers” (HR>1 → higher risk)	Convert to relative risk



Survival Analysis – Python Base

```
from lifelines import KaplanMeierFitter, CoxPHFitter
```

```
df = pd.read_csv("customer_churn.csv")
```

```
kmf = KaplanMeierFitter()
```

```
kmf.fit(durations=df["tenure"], event_observed=df["churned"])
```

```
kmf.plot_survival_function()
```

```
cox = CoxPHFitter()
```

```
cox.fit(df, duration_col='tenure', event_col='churned')
```

```
cox.print_summary()
```

Probable Exercise: Identify which customer segments have the highest churn risk within 6 months.

Model Interpretability & Explainability



SECTION 2: Model Interpretability & Explainability

Why Interpretability Matters?

➤ Interpretability = Bridge between Data Science and Decision Science.

- Black-box models erode stakeholder trust -> Helps **build trust** among executives.
- Regulators demand explainable decisions -> Satisfies **regulatory transparency** (banking, healthcare).
- Empowers storytelling for decision influence -> Thus allows **actionable storytelling**.

Example: A bank rejects a loan → Model must explain *why*.

“Low income and high debt-to-income ratio were key factors.”



SHAP & LIME – Explaining Predictions

SHAP (SHapley Additive exPlanations)

- Based on game theory
- Shows feature contributions to individual predictions

LIME (Local Interpretable Model-Agnostic Explanations)

- Locally approximates model behavior around an observation

Business Example

- SHAP summary identifies marketing spend and income as top churn drivers
- LIME explains individual loan approval cases



Model Interpretability & Explainability – Technique Comparison Table

Method	Type	Scope	Strength	Limitation	Business Fit
SHAP	Model-agnostic	Global & Local	Consistent, game-theoretic	Computationally heavy	Executive dashboards
LIME	Model-agnostic	Local only	Simple, fast	Instability for small changes	Customer-level explainability
Coefficients (OLS/GLM)	Model-specific	Global	Transparent	Only linear	Compliance reporting



Model Interpretability & Explainability – Practical Challenges & Resolutions

Challenge	Issue	Resolution
Computation	SHAP on deep models is slow	Use KernelSHAP / sampling
Stakeholder Understanding	Outputs too technical	Simplify to “Top 3 Drivers per Customer”
Scaling	Explaining millions of predictions	Use global SHAP summary plots



SHAP – Python Base

```
import shap
import xgboost as xgb

model = xgb.XGBRegressor().fit(X_train, y_train)
explainer = shap.Explainer(model, X_train)
shap_values = explainer(X_test)
```

```
# Global importance
shap.summary_plot(shap_values, X_test)
```

```
# Local explanation
shap.plots.waterfall(shap_values[0])
```

Exercise: Identify which customer segments have the highest churn risk within 6 months.

SESSION 3: Neural Networks for Regression & Integration (Dr. Mano)



Advanced Predictive Analytics – Key Technique Comparison Table

Model	Complexity	Interpretability	Overfitting Risk	Business Fit	Tools
Linear Regression	Low	High	Low	KPI forecasting	statsmodels
Regularized Regression	Medium	Medium	Moderate	High-dim data	sklearn
Neural Network	High	Low	High	Complex non-linear	Keras/TensorFlow

SESSION 4: Integration & Storytelling (Mini Project)



Case: Predicting Customer Lifetime Value & Churn

- **Business Problem:** A retail firm wants to predict: (1) Time to churn (Survival Analysis); (2) Expected customer spend (Regularized Regression); (3) Feature influence (SHAP)

Solutions steps:

1. Data Preparation & Feature Scalling
2. Use **Ridge regression** for spend prediction.
3. Use **Cox PH model** for churn time.
4. Apply **SHAP** for interpretability.
5. Create & Present insights as business story:
“High-income but low-engagement customers are 2.3x more likely to churn within 4 months.”

Deliverables Pack (Recommended)

- **Deck (PPT, ~25 slides):** Visuals for methods, comparison charts, business insights.
- **Python Lab (.ipynb):** Hands-on notebook with dataset links (Telco churn, marketing spend).
- **Case Study Worksheet (.docx):** Interpretation and storytelling exercises.

Suggested Slide Organization Summary

Section	Slide Count	Key Content
Overview & Motivation	5	Business problem framing
Regularization	15	Theory + Comparison + Lab
GLMs	10	Poisson/Gamma + Business fit
Survival Analysis	15	Concept + Practical challenges
Model Interpretability	15	SHAP/LIME comparisons
Neural Networks	10	Architecture + Comparisons
Mini Project	10	Integration + Storytelling

Deliverables Tips:

- ✓ Python notebook results
- ✓ 1-slide executive summary per model
- ✓ Stakeholder explanation (non-technical)



Thank You



Key Reference Materials

- Python for Data Science Cheat Sheet: [ds-cheatsheets/Python/Datacamp/img/jupyter-1.png at master · FavioVazquez/ds-cheatsheets · GitHub](https://github.com/FavioVazquez/ds-cheatsheets/blob/master/img/jupyter-1.png)

- Bayesian Statistics: From Concept to Data Analysis (12 hours)
Best to refresh the concepts and learn how Regression analysis is undertaken in terms of statistical validation



This course begins with the basics of probability and Bayes' theorem.



<https://lnkd.in/gBuqigem>

- [How to Pretty-Print Tables in Python | LearnPython.com](https://www.learnpython.com/en/How-to-Pretty-Print-Tables-in-Python)
- <https://www.youtube.com/watch?v=dPwLUkSHLo&list=PL5-da3qGB5ICCsgW1MxlZ0Hq8LL5U3u9y&index=40>